1.8 Reaction Kinetics

Question Paper

Course	CIE A Level Chemistry
Section	1. Physical Chemistry
Topic	1.8 Reaction Kinetics
Difficulty	Medium

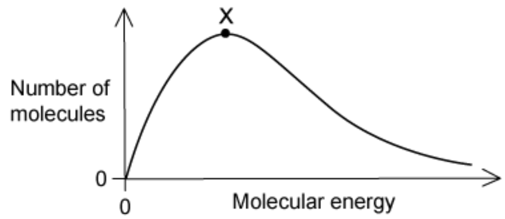
Time allowed: 20

Score: /9

Percentage: /100

Question 1

The distribution of molecular energies in a sample of a gas at a given temperature is shown by the Boltzmann distribution graph below.



If the temperature is increased, what will happen to the position of point X?

A. Fewer molecules possess the most probable energy value so X will shift to the right.

B. Fewer molecules possess the most probable energy value so X will shift to the left.

C. More molecules possess the most probable energy value so X will shift to the left.

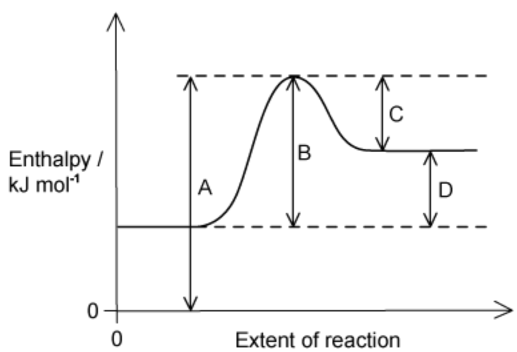
D. The position of X will stay the same but the area under the distribution curve increases.

[1 mark]

Question 2

The diagram shows a reaction pathway for an endothermic reaction.

Which arrow represents the activation energy for the forward reaction?



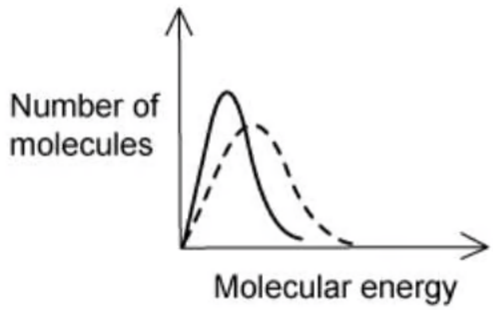
[1 mark]

Question 3

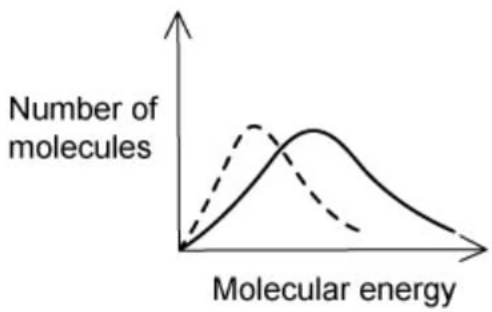
The dotted-line curve on each graph below represents the corresponding distribution for a gas at 300 K.

Which solid-line curve most accurately represents the distribution of molecular energies in the same gas at 500 K?

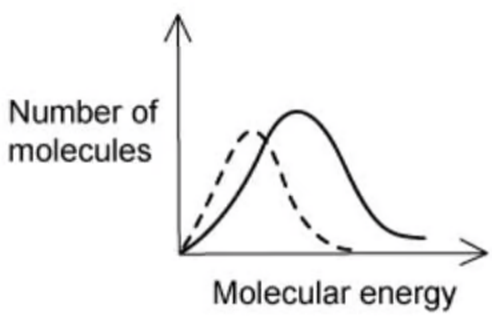
A.



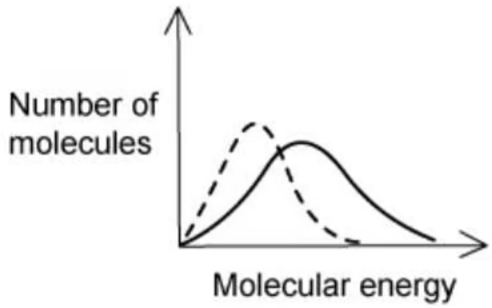
B.



C.



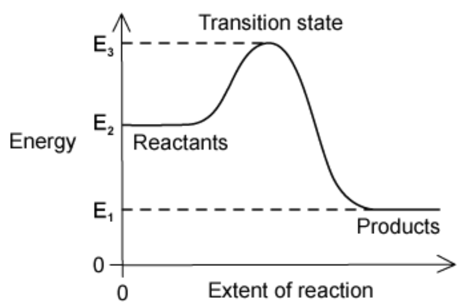
D.



[1 mark]

Question 4

The energies of the reactants, the products and the transition state of a reaction are shown in the reaction pathway diagram below.



Which expression correctly represents how to calculate the activation energy of the forward reaction?

A. $E_1 - E_2$

B. $E_2 - E_1$

C. $E_2 - E_3$

D. $E_3 - E_2$

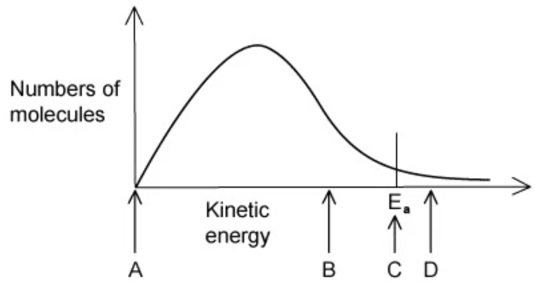
[1 mark]

Question 5

The diagram below represents, for a given temperature, the Boltzmann distribution of the kinetic energy of the molecules in a mixture of two gases that react slowly together without a catalyst.

The activation energy for the reaction, E_a, is marked for the uncatalysed reaction.

What would the position of E_a be if the reaction took place with an effective catalyst?



[1 mark]

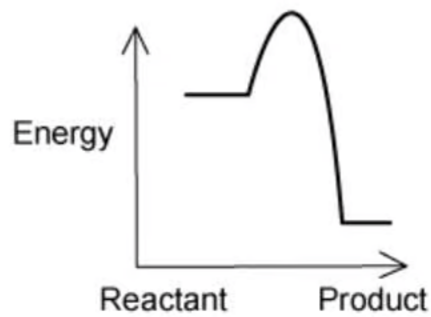
Question 6

Four possible reactions (A, B, C and D) of the equation below are measured at the same temperature.

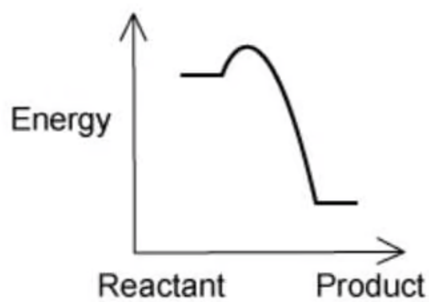
$$X(g) + Y(g) \rightarrow Z(g)$$

Which reaction pathway diagram shows the reaction that would proceed most rapidly and with good yield?

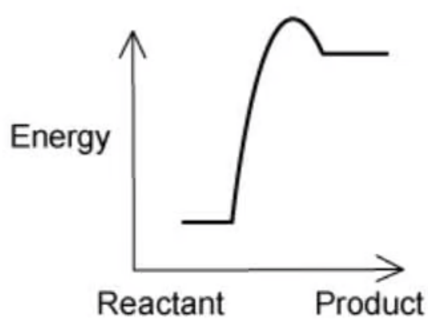
A.



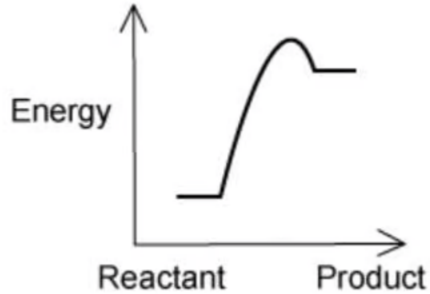
B.



C.



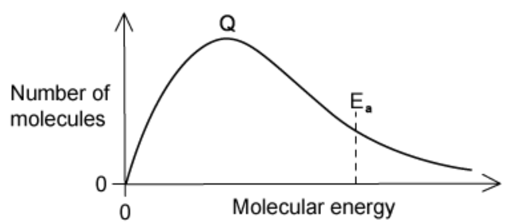
D.



[1 mark]

Question 7

The diagram shows a Boltzmann distribution of molecular energies for a gaseous mixture. The distribution has a peak, labelled Q on the diagram.



What happens when an effective catalyst is added to the mixture?

A. The height of the peak remains the same, and the activation energy moves to the left.

B. The height of the peak decreases and the activation energy moves to the left.

C. The height of the peak remains the same, and the activation energy moves to the right.

D. The height of the peak decreases and the activation energy moves to the right.

[1 mark]

Question 8

A student performs two reactions and measures the rate of product formation.

Reaction 1: 1.5 g of solid calcium carbonate is added to 100 cm^3 of 0.5 M hydrochloric acid.

Reaction 2: 100 cm^3 of distilled water is then added to 100 cm^3 of 0.5 M hydrochloric acid then 1.5 g of solid calcium carbonate is added.

The rate of reaction 1 was faster than the rate of reaction 2.

Which of the following 3 hypotheses correctly describes the difference in the rate?

1 Adding water reduces the frequency of collisions between reactant molecules.

2 Adding water reduces the proportion of effective collisions between reactant molecules.

3 Adding water reduces the proportion of reactant molecules possessing the activation energy.

A. 1 only

B. 1 and 2

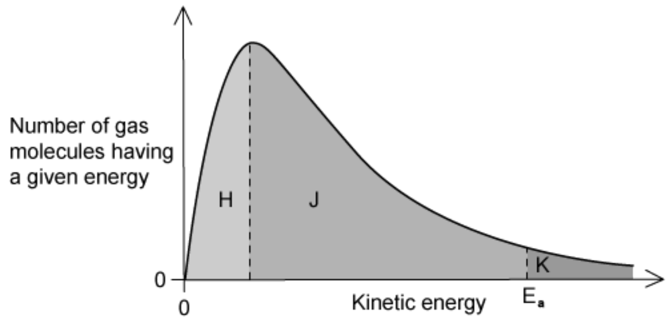
C. 3 only

D. 1, 2 and 3

[1 mark]

Question 9

The Boltzmann distribution shows the number of molecules that have particular kinetic energy at a constant temperature.



If the temperature is decreased by 10 °C, what happens to the size of the areas labelled *H*, *J* and *K*?

	H	***J***	***K***
A	Decreases	Decreases	Decreases
B	Decreases	Increases	Decreases
C	Increases	Decreases	Decreases
D	Increases	Decreases	Increases

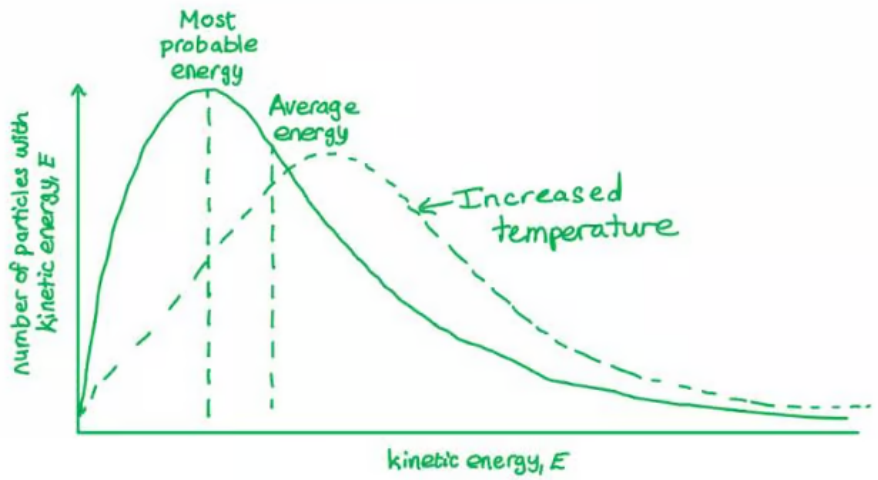
[1 mark]

Model Answers: Medium

1

The correct answer is **A** because:

- The Boltzmann distribution curve shows the peak as the most probable energy.



- When the temperature is increased the number of particles with enough energy for a successful collision is increased.
    - Shifting the peak of the graph down and to the right.

B is incorrect as the graph would shift to the left when the particles have less energy and therefore further away from having the activation energy.

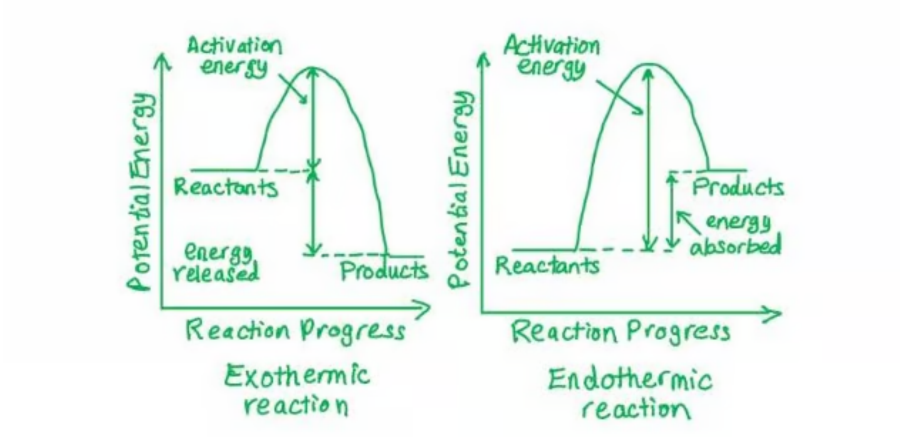
C is incorrect as if more particles have the most probable energy, then there will be fewer particles with the activation energy required for the reaction to take place.

D is incorrect as the area under the graph is the number of particles, the area under the graph would only change if the number of particles changed.

2

The correct answer is **B** because:

- The activation energy in an endothermic reaction is the energy needed for the reactants to react.
- This is the energy from the reactants to the 'hump' in the line as shown in the diagram below.



A is incorrect as this does not correspond to the activation energy.

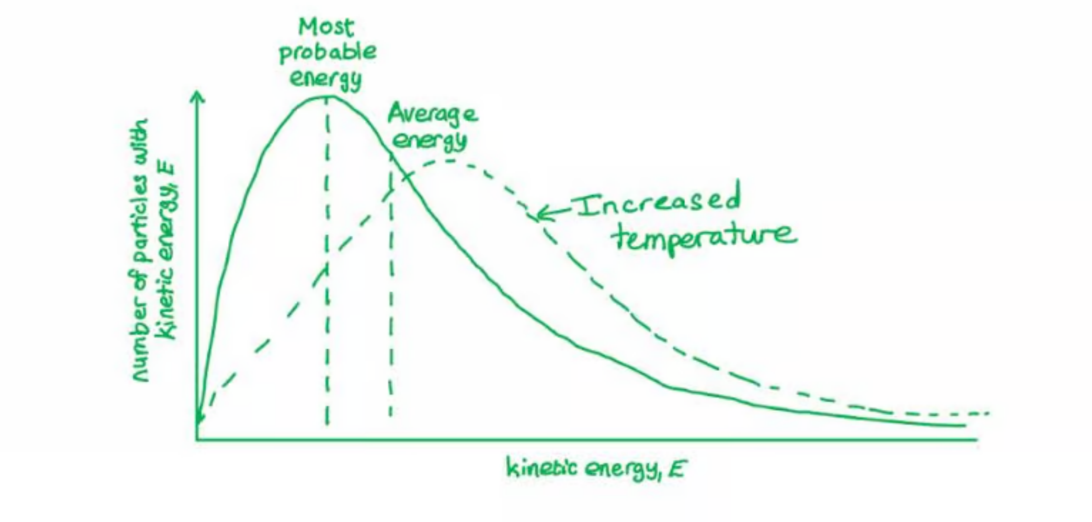
C is incorrect as this is the activation energy for the reverse reaction.

D is incorrect as this is the ΔH of the forward reaction.

3

The correct answer is **D** because:

- When the temperature is increased the number of particles with enough energy for a successful collision is increased.
    - Shifting the peak of the graph down and to the right.



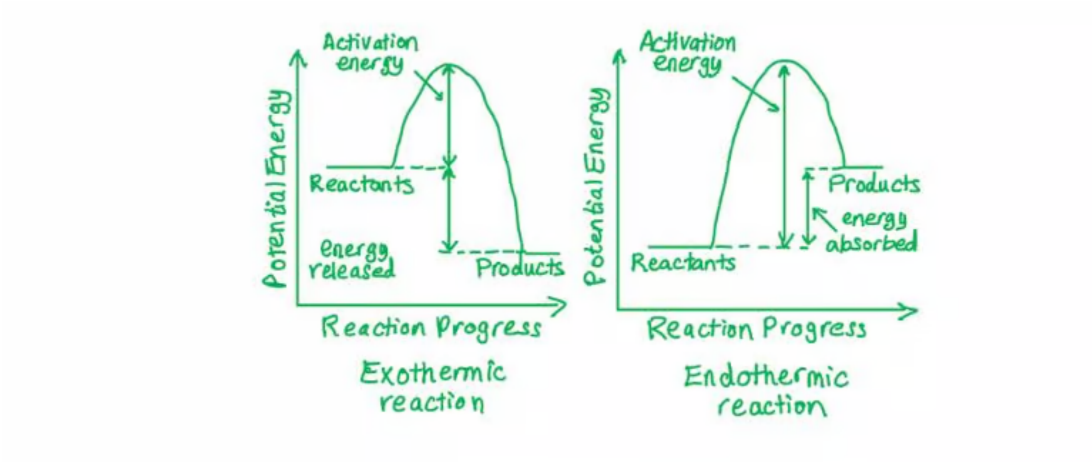
- The graph that matches this profile is **D.**

A, B & **C** are incorrect as all the other graphs do not move down and to the right.

4

The correct answer is **D** because:

- The activation energy is minimum energy particles must possess to break bonds to start a chemical reaction.



- This is the energy from the reactants to the 'hump' in the line; therefore $E_3 - E_2$.

A & C is incorrect as this does not correspond to the activation energy.

B is incorrect as this is the ΔH of the forward reaction.

5

The correct answer is **B** because:

- This is the energy from the reactants to the 'hump' in the line; therefore $E_3 - E_2$.

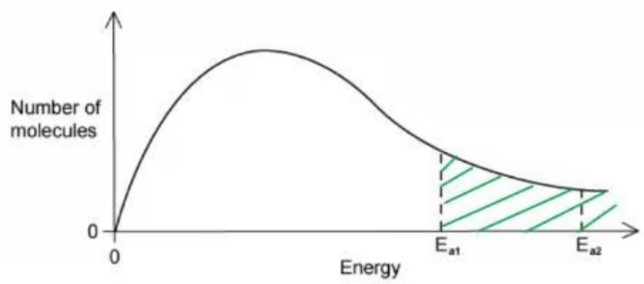
A & C is incorrect as this does not correspond to the activation energy.

B is incorrect as this is the ΔH of the forward reaction.

5

The correct answer is **B** because:

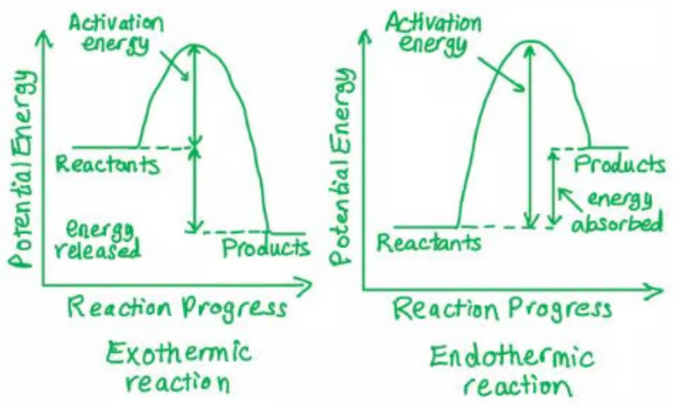
- The definition of a catalyst is that it **lowers the activation energy** by providing an **alternative pathway**; it does not form part of the products and is therefore not used in the reaction.
- Therefore, more particles will have enough energy to react as shown in the shaded area in the graph below.



6

The correct answer is **B** because:

- The question states that the graph must show the reaction that would proceed most rapidly and with the highest yield.
- All the reactions are carried out at the same temperature.
- We are looking for the graph with the smallest activation energy.
  - The diagrams below show energy profiles for an endothermic and exothermic reaction.



- The exothermic graph, which is thermodynamically favourable, with the lowest activation energy is **B.**

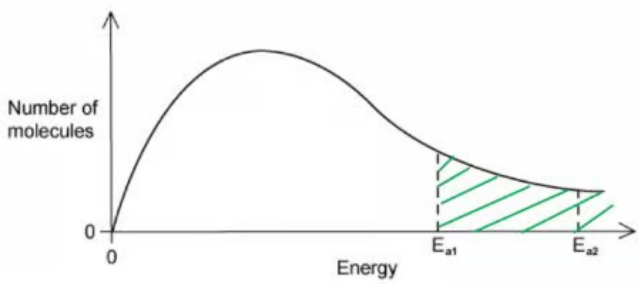
A is incorrect as it has higher activation energy than B.

C & **D** are incorrect as they are both endothermic reactions and thermodynamically unfavoured.

7

The correct answer is **A** because:

- The definition of a catalyst is that it **lowers the activation energy** by providing an **alternative pathway**; it does not form part of the products and is therefore not used in the reaction.



- A catalyst does not change the number of particles at each energy so the peak of the graph will stay the same.
- The activation energy moves to the left as a catalyst offers an alternative pathway.

B is incorrect as the height of the peak does not change.

C is incorrect as if the activation energy is moved to the right, then fewer particles would have the activation energy, and the rate of the reaction would go down.

D is incorrect as the height of the peak does not change. If the activation energy is moved to the right, then fewer particles would have the activation energy, and the rate of the reaction would go down.

8

The correct answer is **A** because:

- When the water is added in reaction 2 this dilutes the concentration of the hydrochloric acid.
- When the concentration is diluted, there are less frequent collisions between reacting particles reducing the rate.

B & **D** are incorrect as the effectiveness of the collision is not determined by the concentration, but the orientation and the energy of the colliding particles.

C is incorrect as diluting the concentration does not affect the activation energy.

9

The correct answer is **C** because:

- As the temperature is decreased the line shifts to the left and the peak upwards.
- This would mean a greater number of particles with less energy, so the area H increases and the other two areas decrease.